

National University of Computer and Emerging Sciences
Lahore Campus

Programming Fundamentals

Deadline: September 10, 2025

Course Instructor

Ms. Anosha Khan

Course Teacher Assistant

Muhammad Annus Shahzad

Assignment 1

Total Time (Days): 7

Total Marks: 100

Total Questions: 8

Instructions to be followed

Include comments in your code wherever necessary to improve clarity. The use of Artificial Intelligence tools or Large Language Models (LLMs), such as ChatGPT or DeepSeek, is strictly forbidden. Any student found using AI or LLMs to complete this assignment will receive **zero marks** and may face **serious disciplinary action (DC)** in the future.

If you are unsure about any part of the assignment, consult your **TA for clarification** before making assumptions.

Submission Instructions

Solve each question in a **separate C++ file**. Name each file using this format (RollNumber_Q1.cpp, RollNumber_Q2.cpp) Example: 25L_XXXX_Q1.cpp, 25_XXXX_Q2.cpp and so on. Put all your C++ files into **one ZIP file** with the following format (**25L_XXXX.zip**) and submit that **ZIP File** on Google Classroom.

Important: Submissions that do not follow these instructions will result in **deduction of marks**. It is your duty to submit working code on GCR following the format given above. Students are expected to ensure that their programs **handle edge cases** with a user-friendly output if needed. Also handle **unwanted values & errors**.

The syllabus includes Selection Structures (if, else if, else, and ternary operators) and topics covered before this. The use of advanced topics will lead to deduction of marks.

Flip the Page for Questions

National University of Computer and Emerging Sciences

Lahore Campus

Question 01

Create a Program that takes 5 Numbers (Integers) from User.
Print them in ascending and descending order

Question 02

Create a program that calculates the area and perimeter of a square, rectangle, and circle. The program should also calculate only the perimeter of a triangle, determine the type of triangle (Equilateral, Isosceles, or Scalene) and check whether a given rectangle is a square

Question 03

Suppose your semester is about to end, you know your Total Absolutes. Write a Program to calculate your Grade based on FAST NUCES Absolute Grading Scheme. Also Print out the Relevant Points (A+ 4.00, A 4.00, A- 3.67 ...) along with it

Question 04

Write a program that takes the grade points and credit hours for N courses as input, then calculates and displays the SGPA (Semester Grade Point Average).

$$\frac{\sum_{i=0}^N (Grade\ Point)^i \times (Credit\ Hours)^i}{\sum_{i=0}^N (Credit\ Hours)^i}$$

- Here N is the number of courses offered to you people in 1st Semester and **is constant**.
 - $(Grade\ Point)^i$ represent the expected Grade points in ith Course (A+ 4.00, A 4.00, A- 3.67 ...)
 - $(Credit\ Hours)^i$ represent the Credit Hours of ith Course (3,2,1)
-

Question 05

Write a program that takes a **5-digit zip code** from the user and tells which **city & province** it belongs to.

For example:

- Codes starting with 75 → Karachi
- Codes starting with 54 → Lahore
- Codes starting with 51 → Islamabad
- Codes starting with 42 → Faisalabad

And so on. Your Program should be able to identify all provinces. And Maximum of 10 Cities in Total.

National University of Computer and Emerging Sciences

Lahore Campus

Question 06

A coffee shop named “**Caffeine Hub**” sells three types of coffee: **Espresso**, **Cappuccino**, and **Latte**. The prices are as follows:

- Espresso = Rs. 150
- Cappuccino = Rs. 180
- Latte = Rs. 200

The user will enter the type of coffee they want and the number of cups. Also, ask if the customer wants **extra sugar (Rs. 20)** or **whipped cream (Rs. 30)**. Finally, calculate the total bill with **10% service tax**, and display the complete order details.

Question 07

Write a program that takes a **8-digit number** from the user and prints:

- The number in **reverse** order
- The **sum** of the digits
- Check If the Number is a palindrome

Question 08

Write a program that takes a character from user and prints:

- A Number if the Character is a Number. (0-9)
- Lowercase Alphabet if it is in small-caps. (a-z)
- Uppercase Alphabet if it is in uppercase. (A-Z)
- Punctuation marks if it is a Punctuation mark. (.,?";:~`)
- Brackets/Parenthesis if it is. (){}[]
- Symbols if it belongs to Symbols (_-+*&^%\$#@!~<>\/)
- Space If it is a Space
- Special Character, (A Character that can not be displayed on computer screen)

I hope you all will learn something new
while doing this Assignment
Best of Luck Guys

See you all energetic than before in Evaluation
Best Wishes